

CYBERSYN COUCH

DUAL-JOYSTICK • MOBILE • OPEN
HARDWARE

Two-seater cybernetic couch on wheels.
Inspired by Chile's 1971 Operations Room.
Every button does something.

V0.1 — MARCH 2026





Dual Joystick Control

Each armrest houses a hall-effect 2-axis joystick and 4 Cherry MX mechanical buttons.

Joystick	Hall-effect, +/-25 deg, spring-return
Shaft	120mm anodized aluminum, orange grip
Buttons	4x Cherry MX per arm (light, scene, music, FN)
Polling	50 Hz, 16-bit ADC, 8% deadzone
Feedback	Haptic motor + LED ring per armrest

5 Control Modes

Press SWAP to propose • Partner confirms or you force-swap • 3-second timeout

1

Independent

Left stick → left side
Right stick → right side

Default. Full sovereignty.

2

Swapped

Left stick → right side
Right stick → left side

Cross-control.

3

Merged

Left stick = X axis
Right stick = Y axis

Cooperative 2D input.

4

Pilot / Co-pilot

Pilot has full control
Co-pilot override >80%

Asymmetric command.

5

Custom

User-defined mapping via YAML config
Fully programmable.

Lighting: X = color temp, Y = brightness, push = zone on/off

Music: X = track skip, Y = volume, push = play/pause

Fun Mode: X = steer spaceship, Y = throttle, push = fire lasers!

Center Console

Display	1.3" OLED 128×64 – mode, track, status
Row 1	LIGHTS • VOLUME • MUSIC • POWER
Row 2	SWAP • NFC reader • MODE SELECT
Row 3	RANDOM • PARTY • AMBIENT • SECRET

NFC profiles – tap a card to load your lighting scenes, music prefs, and joystick curves

SWAP button – guarded mechanical switch triggers cooperative handshake



I/O At a Glance

4

analog axes

2 per joystick, 16-bit ADC

20

buttons

8 FN + 8 dashboard + 2 push + 2 rotary

125

RGB LEDs

WS2812B, 4 zones: arms, base, back

3

haptic motors

DRV2605L + LRA per zone

1-3

OLED displays

SH1106 128×64 green-on-black

1

NFC reader

RC522 user profile cards

2

speakers

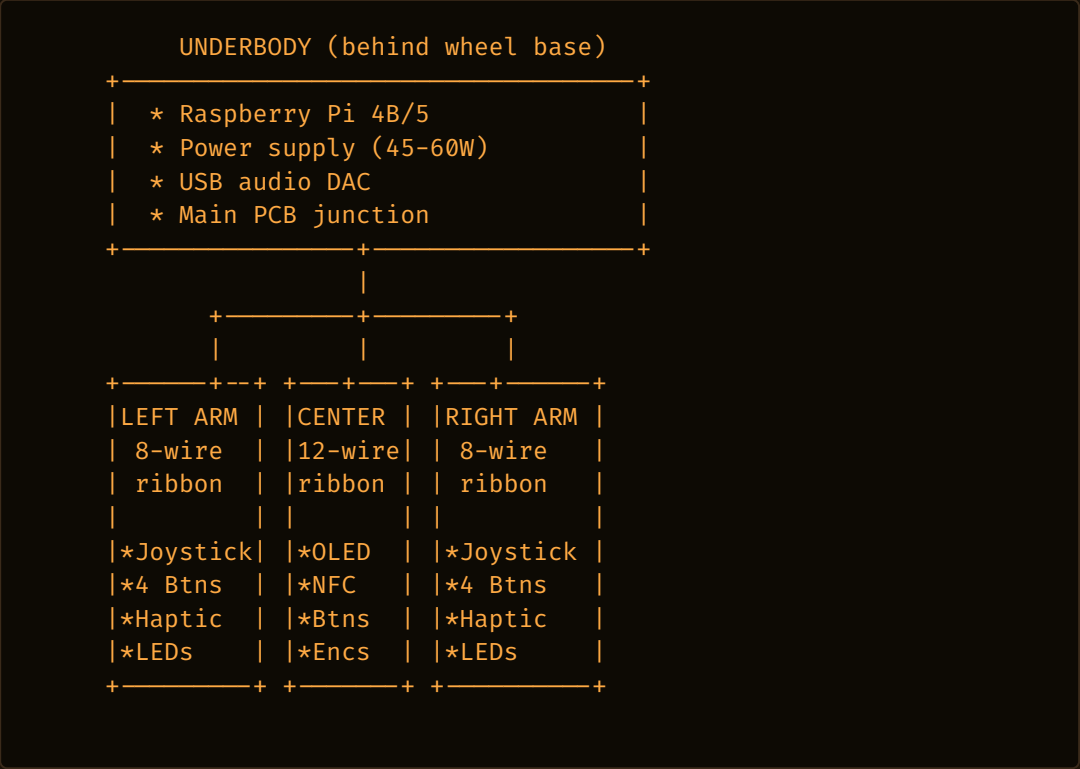
5W stereo via USB DAC

4

caster wheels

Locking, 75mm, 200kg rated

Electronics Architecture



COMPONENT	PART	BUS
Brain	Raspberry Pi 4B/5	Host
ADC	ADS1115 16-bit	I2C 0x48
I/O x2	MCP23017	I2C 0x20-21
I2C Mux	TCA9548A	I2C 0x70
Haptic x3	DRV2605L + LRA	I2C (mux)
Display	SH1106 OLED	I2C 0x3C
NFC	RC522	SPI
LEDs	WS2812B x125	GPIO
Audio	USB DAC + 2x5W	USB

Software: Python 3.11 asyncio, MQTT event bus, Home Assistant integration, NFC user profiles

A cinematic background image showing two men sitting on a futuristic, white, pod-like sofa with orange interior lighting. They are in a high-rise apartment with floor-to-ceiling windows overlooking a vibrant city skyline at night, featuring illuminated skyscrapers like the Chrysler Building. The man on the left is wearing a brown leather jacket and dark pants, while the man on the right is wearing a brown jacket and dark pants, holding a small object in his hand. The overall mood is sophisticated and futuristic.

10 Easter Eggs Built In

Party Mode

Deep Space

Konami Code

Algedonic Alert

Secret DJ

Cybersyn Boot

Random Wisdom

Mood Sync

Chilean Flag

Ghost in Machine

Custom easter eggs definable via YAML – button sequences, LED patterns, audio, haptic, display text

Build Plan & Cost

PHASE	WHAT	TIME
0	Breadboard PoC – all sensors, validate I2C, test swap	2-3 wk
1	Custom PCB – KiCad schematic, JLCPCB fab	2-3 wk
2	3D printed armrest mockup with live electronics	2-3 wk
3	Fiberglass shell – foam plug, mold, cable channels	4-6 wk
4	Wheel base – steel frame, 4x locking casters, 200kg rated	1-2 wk
5	Upholstery + final assembly	2-3 wk
6	Software tuning – calibration, scenes, easter eggs	2-3 wk
7	Multi-couch MQTT network – voting, Opsroom mode	Future

\$1,350

estimated – first unit

~\$975

per unit at 2+ (mold reuse)

15-21

weeks to first couch

Physical Specifications

1800mm

overall width
71" end to end

850mm

backrest height
33.5" ergonomic

45-55

kg total weight
Fiberglass monocoque

200

kg load rating
Steel wheel frame



Shell	White fiberglass monocoque, compound curves
Upholstery	Burnt orange (Pantone 1665 C)
Base	Steel frame + 4x 75mm locking caster wheels
Armrests	200mm wide, recessed controls
Center	250mm console between seats

CYBERSYN COUCH

OPEN HARDWARE • OPEN SOFTWARE • OPEN FUTURE

gitea.jeffemmett.com/jeff/cybersyn-chair

Full spec: `docs/cybersyn-couch-spec.md`

Hardware: CERN-OHL-S-2.0 • Software: MIT • Docs: CC BY-SA 4.0

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